

Package ‘carbonpredict’

September 24, 2025

Title Predict Carbon Emissions for UK SMEs

Version 1.0.0

Description Predict scope 1, 2 and 3 carbon emissions using pretrained models for UK SMEs as well as predicting scope 3 emissions hotspots.

License MIT + file LICENSE

Encoding UTF-8

Roxygen list(markdown = TRUE)

RoxygenNote 7.3.2

Imports dplyr,
ggplot2,
progress,
lmerTest,
networkD3

Suggests testthat (>= 3.0.0),
mockery,
png,
grid

Config/testthat/edition 3

Contents

batch_predict_emissions	2
batch_sme_plots	2
plot_scope3_emissions	3
plot_sme_emissions	4
sme_emissions_profile	4
sme_scope1	5
sme_scope2	6
sme_scope3	6
sme_scope3_hotspots	7
Index	8

batch_predict_emissions

Batch Predict Emissions

Description

Prediction entry point for batch SME and agriculture emissions

Usage

```
batch_predict_emissions(data, output_path = NULL, company_type = "sme")
```

Arguments

data	A single entry (list or named vector), a data frame, or a path to a CSV file. The data should contain company_name, 2-digit UK sic_code, and annual turnover columns.
output_path	Optional file path to save the results as a CSV. If NULL, results are not saved to a file.
company_type	A single parameter "sme" or "farm" to determine which emission prediction funtions to call (defaults to "sme").

Value

A data frame with input columns and predicted emissions for each scope (in tCo2e). Optionally saved to a CSV file.

Examples

```
sample_data <- read.csv(system.file("extdata", "sme_examples.csv", package = "carbonpredict"))
sample_data <- head(sample_data, 3)
batch_predict_emissions(data = sample_data, output_path = NULL, company_type = "sme")
```

batch_sme_plots

Batch SME Plots

Description

Batch plot SME Scope 1 & 2 emissions

Usage

```
batch_sme_plots(data, output_path = NULL)
```

Arguments

data	A data frame or path to a CSV file with columns "sic_code", "turnover", and optionally "company_name".
output_path	Optional directory to save plots. If NULL, plots are not saved.

Value

Donut chart plots showing scope 1 and 2 predicted emissions (in tCo2e) for each row in the data. Optionally saved to a directory as PNG files.

Examples

```
sample_data <- read.csv(system.file("extdata", "sme_examples.csv", package = "carbonpredict"))
sample_data <- head(sample_data, 3)
batch_sme_emissions <- batch_predict_emissions(
  data = sample_data,
  company_type = "sme",
  output_path = NULL)
batch_sme_plots(data = batch_sme_emissions, output_path = NULL)
```

plot_scope3_emissions *Plot Scope 3 Emissions Breakdown*

Description

Plots a Sankey diagram showing the breakdown of Scope 3 emissions by category.

Usage

```
plot_scope3_emissions(scope3_df, company_name = NULL)
```

Arguments

scope3_df	Data frame output from sme_scope3 (must contain 'Category', 'Description', and 'Predicted Emissions (tCO2e)').
company_name	Optional company name to include in the chart title (character string).

Value

A Sankey plot showing a breakdown for predicted emissions of each Scope 3 category.

Examples

```
scope3_df <- sme_scope3(85, 12000000)
plot_scope3_emissions(scope3_df, company_name = "ABC")
```

plot_sme_emissions	<i>Plot SME Emissions</i>
--------------------	---------------------------

Description

Plot a donut chart of Scope 1,2 and 3 emissions

Usage

```
plot_sme_emissions(
  scope1_emissions,
  scope2_emissions,
  scope3_emissions,
  company_name = NULL
)
```

Arguments

scope1_emissions	Value for total Scope 1 emissions (numeric).
scope2_emissions	Value for total Scope 2 emissions (numeric).
scope3_emissions	Value for total Scope 3 emissions (numeric).
company_name	Optional company name to include in the chart title (character string).

Value

A ggplot2 donut chart showing predicted emissions for each scope.

Examples

```
scope_1 = sme_scope1(85, 12000000)
scope_2 = sme_scope2(85, 12000000)
scope_3 = sme_scope3(85, 12000000)
plot_sme_emissions(
  scope1_emissions = scope_1$`Predicted Emissions (tCO2e)`,
  scope2_emissions = scope_2$`Predicted Emissions (tCO2e)`,
  scope3_emissions = scope_3[scope_3$Category == "Total", "Predicted Emissions (tCO2e)"][[1]],
  company_name = "ABC")
```

sme_emissions_profile	<i>SME Emissions Profile</i>
-----------------------	------------------------------

Description

Calls the Scope 1, 2 and 3 emissions prediction functions and returns their results as a list and plots a donut chart

Usage

```
sme_emissions_profile(sic_code, turnover, company_name = NULL)
```

Arguments

sic_code	A 2-digit UK SIC code (numeric).
turnover	Annual turnover value (numeric).
company_name	Optional company name for labeling plots (character string).

Value

A list with four elements: `scope1`, `scope2`, `scope3`, `scope3_hotspots`, each containing the predicted carbon emissions data frame (in tCo2e), the top 5 scope 3 emissions hotspots, as well as a donut chart and Sankey diagram showing the emissions breakdowns.

Examples

```
sme_emissions_profile(sic_code = 85, turnover = 12000000, company_name = "ABC")
```

`sme_scope1`*Predict SME Scope 1 Emissions*

Description

This function loads a pre-trained emission model to predict scope 1 carbon emissions for a given SIC code and turnover.

Usage

```
sme_scope1(sic_code, turnover)
```

Arguments

sic_code	A 2-digit UK SIC code (numeric).
turnover	Annual turnover value (numeric).

Value

A data frame with predicted emissions (in tCo2e).

Examples

```
sme_scope1(sic_code = 85, turnover = 12000000)
```

sme_scope2	<i>Predict SME Scope 2 Emissions</i>
------------	--------------------------------------

Description

This function loads a pre-trained emission model to predict scope 2 carbon emissions for a given SIC code and turnover.

Usage

```
sme_scope2(sic_code, turnover)
```

Arguments

sic_code	A 2-digit UK SIC code (numeric).
turnover	Annual turnover value (numeric).

Value

A data frame with predicted emissions (in tCo2e).

Examples

```
sme_scope2(sic_code = 85, turnover = 12000000)
```

sme_scope3	<i>Predict SME Scope 3 Emissions</i>
------------	--------------------------------------

Description

This function loads pre-trained emissions models to predict scope 3 carbon emissions for a given SIC code and turnover.

Usage

```
sme_scope3(sic_code, turnover)
```

Arguments

sic_code	A 2-digit UK SIC code (numeric).
turnover	Annual turnover value (numeric).

Value

A data frame with predicted emissions (in tCo2e) for each scope 3 category.

Examples

```
sme_scope3(sic_code = 85, turnover = 12000000)
```

sme_scope3_hotspots	<i>Predict Top 5 SME Scope 3 Emissions Hotspots</i>
---------------------	---

Description

This function uses pre-computed results to predict the top 5 scope 3 carbon emissions hotspots for a given SIC code.

Usage

```
sme_scope3_hotspots(sic_code)
```

Arguments

sic_code	A 2-digit UK SIC code (numeric).
----------	----------------------------------

Value

A data frame with the top 5 emissions hostposts for scope 3.

Examples

```
sme_scope3_hotspots(sic_code = 85)
```

Index

`batch_predict_emissions`, [2](#)
`batch_sme_plots`, [2](#)

`plot_scope3_emissions`, [3](#)
`plot_sme_emissions`, [4](#)

`sme_emissions_profile`, [4](#)
`sme_scope1`, [5](#)
`sme_scope2`, [6](#)
`sme_scope3`, [6](#)
`sme_scope3_hotspots`, [7](#)